

MUX - 4

Implementation of ~~MUX~~ Boolean function using MUX:-

$f(A, B, C, D) = \sum m(1, 4, 5, 7, 9, 12, 13)$ using 4×1 MUX

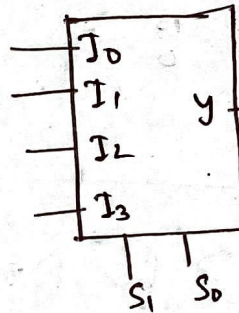
AB \ CD	00	01	11	10
00	0	1	0	0
01	1	1	1	0
11	1	1	0	0
10	0	1	0	0

$$I_0 = \bar{C}D$$

$$\bar{C} + D = I_1$$

$$I_2 = A\bar{B}C$$

$$I_3 = \bar{C}D$$



$$S_0 = B$$

$$S_1 = A$$

A	B	S
0	0	$I_0 = C$
0	1	$I_1 = \bar{C}$
1	0	$I_2 = \bar{C}$
1	1	$I_3 = C$

A	B	C_0
0	0	0
0	1	C
1	0	C
1	1	1

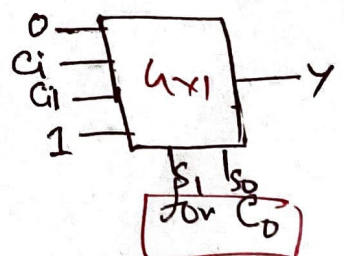
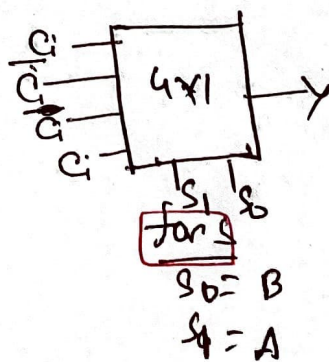
$S_1 S_0$	Y
0 0	$I_0 = \bar{C}D$
0 1	$I_1 = D$
1 0	$I_2 = \bar{C}D$
1 1	$I_3 = C$

1 bit FA using MUX (4x1):-

A	B	C_{in}	S	C_0
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

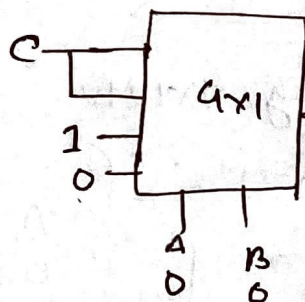
BC _i	for S	
A	0	1
0	0	1
1	1	0

BC _i	for C ₀	
A	0	1
0	0	0
1	0	1



Logical Expression from MUX :

Ex.-1)



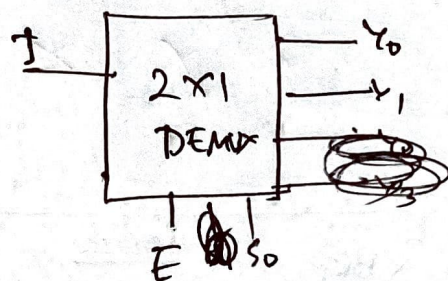
$$Y_1 = \bar{A}\bar{B}C + A\bar{B}C + AB'1 + AB \cdot 0$$

$$= A'C[B' + B] + AB'$$

$$= A'C + AB'$$

DEMUX $\Rightarrow$ 1:2 DEMUX

$\hookrightarrow$ one input $\Rightarrow$ many output



E	S ₀	Y ₀	Y ₁
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

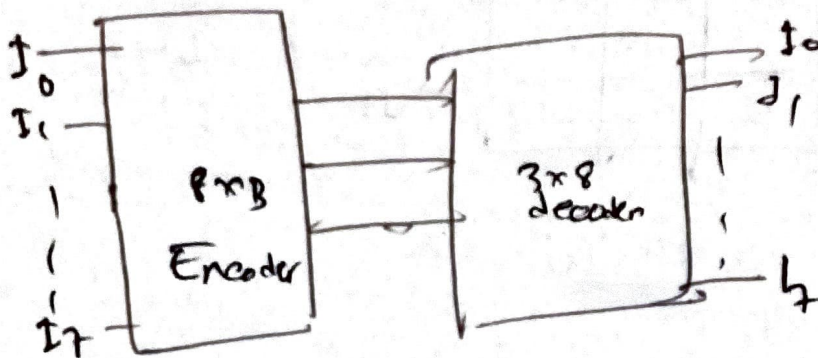
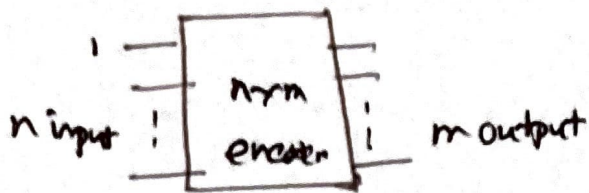
$$Y_0 = E\bar{S}_0$$

$$Y_1 = ES_0$$

Encoder & Decoder:-

Encoder

$\hookrightarrow$ n input, m output



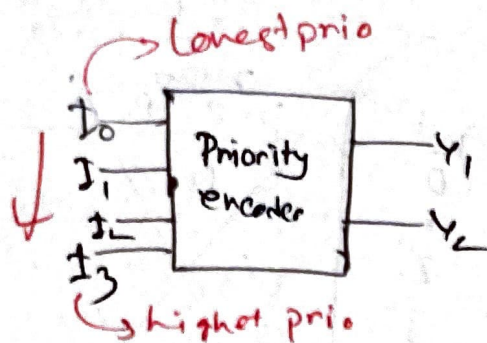
Encoder

Types :-

- Priority encoder
- Decimal to BCD
- Octal to Binary
- Hex to binary

Priority Encoder :-

I_3	I_2	I_1	I_0	Y_1	Y_0
0	0	0	0	x	x
0	0	0	1	0	0
0	0	1	x	0	1
0	1	x	x	1	0
1	x	x	x	1	1



for Y_1

$I_2 I_3$	00	01	11	10
I_1				
00	X	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$Y_1 = I_2 + I_3$$

for Y_0

$I_2 I_3$	00	01	11	10
I_1				
00	X	0	1	1
01	0	0	0	0
11	X	X	X	X
10	1	1	1	1

→ further row $I_2 = 1$
 $I_3 = 1$
 $I_2 \oplus I_3$
 $I_2 \oplus I_3$

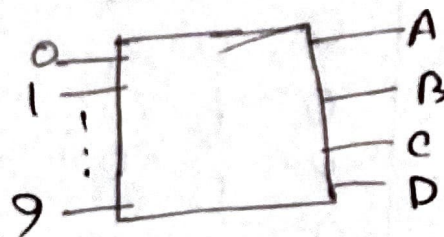
$$Y_0 = I_3 + I_1 \oplus I_2$$

Decimal to BCD encoder

Input	Output	D	C	B	A
0		0	0	0	0
1		0	0	0	1
2		0	0	1	0
3		0	0	1	1
4		0	1	0	0
5		0	1	0	1
6		0	1	1	0
7		0	1	1	1
8		1	0	0	0
9		1	0	0	1

Decimal to BCD Encoder

Input	Output			
	D	C	B	A
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

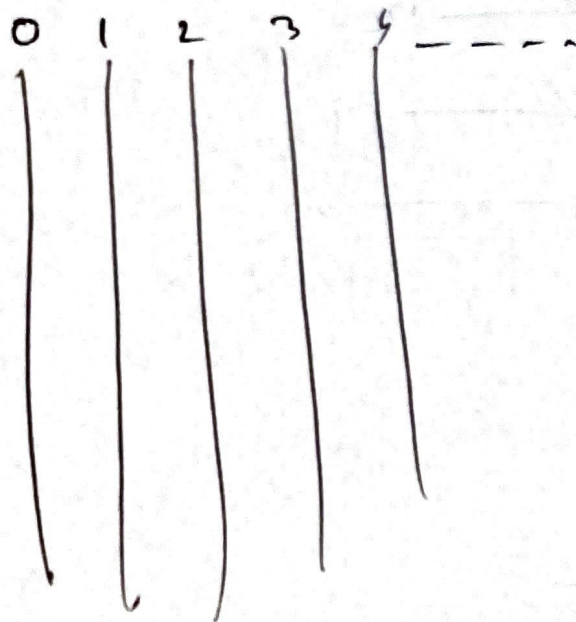


$$D = 8 + 9$$

$$C = 4 + 5 + 6 + 7$$

$$B = 2 + 3 + 6 + 7$$

$$A = 1 + 3 + 5 + 7 + 9$$

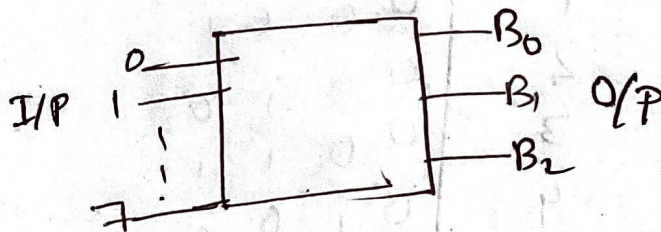


Octal to Binary

Input	B_2	B_1	B_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

$$(61)_8 \rightarrow (???)_2$$

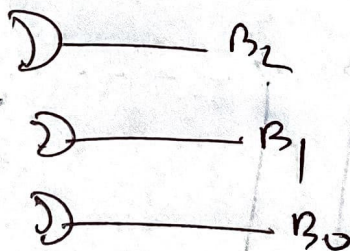
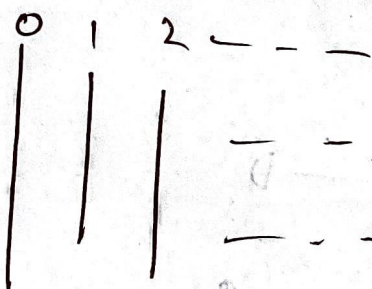
$$\begin{matrix} \swarrow & \downarrow & \searrow \\ (111) & (110) & (001) \end{matrix}_2$$



$$B_2 = 4 + 5 + 6 + 7$$

$$B_1 = 2 + 3 + 6 + 7$$

$$B_0 = 1 + 3 + 5 + 7$$



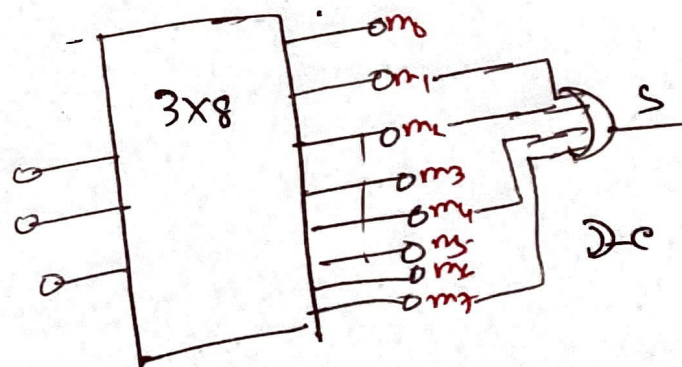
HEX $\rightarrow$ Bin

Same as M01

FA using Decoder :-

T.T.

A	B	C _{in}	S	C _o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$f_{C_0} = \sum(m_3, m_5, m_4, m_7)$$

$$f_S = \sum(m_1, m_2, m_4, m_7)$$